CODEALPHA INTERNSHIP

Iris Flower Classification

Project design, findings, and reproducibility

Prepared for **nisa3131**
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Standard design report layout approved by the user. Findings use the uploaded datasets and measured outputs from the accompanying projects.

Iris Flower Classification

REPOSITORY CodeAlpha_IrisFlowerClassification

1 Objective

Classify setosa, versicolor, and virginica flowers from sepal length, sepal width, petal length, and petal width. The task covers supervised classification, preprocessing, model comparison, and evaluation on held-out examples.

2 Data and provenance

Input: the user-supplied archive.zip, containing Iris.csv. The task document links to https://www.kaggle.com/datasets/saurabh00007/iriscsv. That link is cited as the assignment's dataset reference; these analyses use the uploaded files rather than a new download. Original collection details and redistribution licensing were not independently verified. No dataset license is asserted.

Attribute	Observed data / treatment
Observations	150; 50 per species
Predictors	Four flower measurements in centimetres; Id excluded
Data quality	No missing values or duplicate complete rows
Repeated measurements	3 repeated measurement rows; 147 unique measurement groups
Duplicate treatment	Retain observations; keep identical measurement groups together in every split

Identical measurements can represent separate flowers, so they are retained. Their group identity is used for both the train/test partition and cross-validation to prevent the same measurement vector appearing on both sides. No identical measurement group has conflicting species labels.

3 Method and evaluation

With random seed 42, the first fold from a shuffled five-fold StratifiedGroupKFold creates 120 training and 30 held-out observations. Each held-out species has 10 examples, and no measurement group crosses the partition. Group constraints make stratification approximate in general.

Four candidates are compared: a most-frequent-class baseline, logistic regression, k-nearest neighbours, and a radial-basis support vector classifier. Numeric scaling is inside each estimator's pipeline and therefore fit only on the training portion of each fold. Predefined grids are C = 0.1, 1, 10 for logistic regression and SVM, and k = 3, 5, 7 for neighbours.

Training-only, five-fold group-aware cross-validation selects hyperparameters using mean macro F1. Model-level ties use mean CV accuracy and then model name. The winner is fit on the full training partition and evaluated once on the held-out partition. The test score does not choose the model.

4 Results and interpretation

Candidate	CV macro F1 mean ± SD	CV accuracy
logistic regression	0.9666 ± 0.0312	0.9667
knn	0.9582 ± 0.0264	0.9583
rbf svm	0.9582 ± 0.0264	0.9583
dummy most frequent	0.1667 ± 0.0000	0.3333

Training cross-validation scores. The standard deviation describes variation across five folds, not a confidence interval.

Selected model: logistic regression (C = 10). Its mean CV macro F1 is 0.9666. Held-out accuracy and macro F1 are 1.000 and 1.000, respectively. The most-frequent baseline reaches 0.333 accuracy and 0.167 macro F1.

Species	Test n	Precision	Recall	F1
setosa	10	1.000	1.000	1.000
versicolor	10	1.000	1.000	1.000
virginica	10	1.000	1.000	1.000

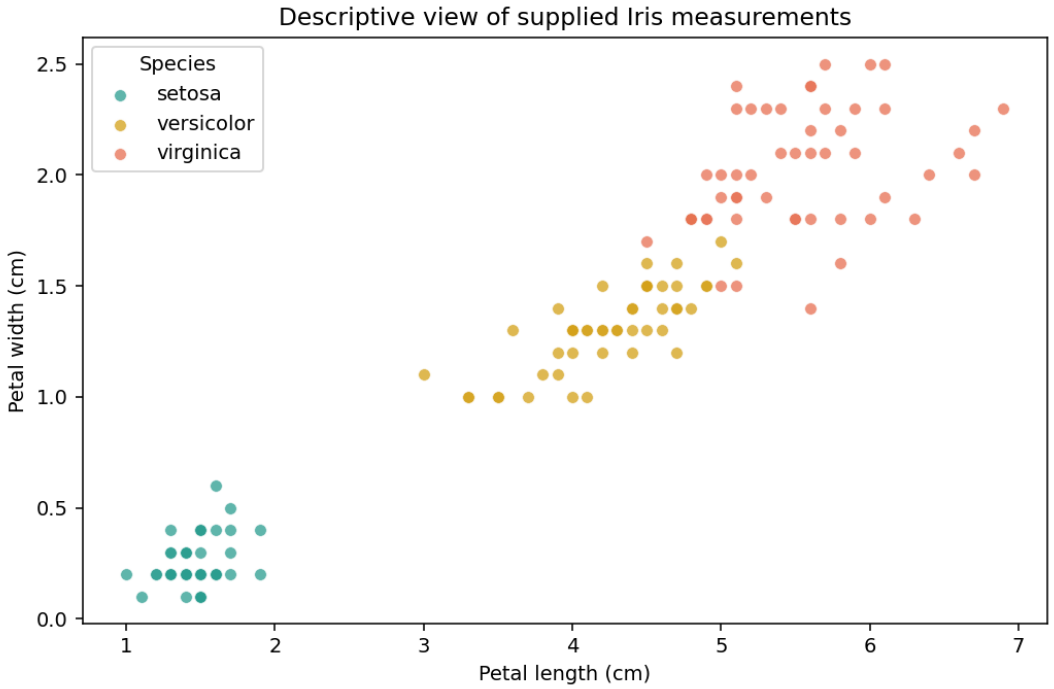


Figure 1. Descriptive petal measurements in the supplied dataset. Setosa is visibly separated; versicolor and virginica overlap more.

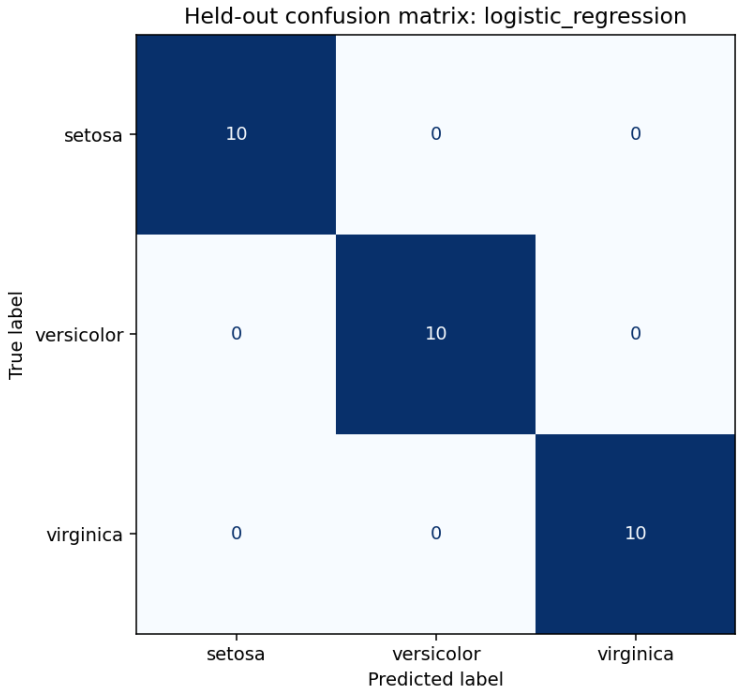


Figure 2. Held-out confusion matrix for the CV-selected model. All 30 test observations are correctly classified in this split.

5 Limitations and next steps

- A perfect score on 30 held-out examples has substantial sampling uncertainty. It does not establish performance on flowers from new locations, instruments, or populations.
- Grouping identical vectors reduces direct duplicate leakage; unrecorded biological or collection relationships may still remain.
- The labels and measurements are not independently verified. This is an educational model, with no external validation or deployment assessment.
- The next evaluation should use an independently collected, well-documented flower dataset and report uncertainty before relying on predictions operationally.

6 Reproducibility

Validation recorded on 19 October 2026: the delivered notebook executed 4 code cells with 0 recorded errors. The analysis pipeline and its built-in data/evaluation checks completed successfully. The project’s 2 unit tests also passed.

Install the pinned requirements and rerun from the repository root. The notebook calls the same analysis pipeline.

```
python -m venv .venv
. .venv/bin/activate
python -m pip install -r requirements.txt
python analysis.py
python -m unittest discover -s tests -v
```

outputs/metrics.json records versions, split sizes, selected parameters, and per-class evaluation. outputs/data_audit.json records the source SHA-256 and quality checks. split_assignments.csv and cv_results.csv expose group membership and every tried configuration; test_predictions.csv records held-out outcomes. The final estimator is fit on training data only.